

Synoptic CB: Porewater Nutrients

May 2024 Samples

2025-12-22

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##Run Information

```
cat("Run Information: NAME ") #lets you know what section you're in
```

Run Information: NAME

```
#set the run date & user name
run_date <- "5/12/2025"
sample_year <- "2024"
sample_month <- "MAY"
user <- "Isabelle Van Benschoten"

##Data entered incorrectly
Old_ID_1 <- "SWH_202405_SWAMP_LysA_10"
New_ID_1 <- "SWH_202405_SWAMP_LysA_10cm"

#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month
NOx_files <- c("Raw Data/COMPASS_Synoptic_CB_202405_VNOx_1.csv",
               "Raw Data/COMPASS_Synoptic_CB_202405_VNOx_2.csv",
               "Raw Data/COMPASS_Synoptic_CB_202405_VNOx_4.csv")
NH3_PO4_files <- c("Raw Data/COMPASS_Synoptic_CB_202405_NH3_PO4_1.csv",
                   "Raw Data/COMPASS_Synoptic_CB_202405_NH3_PO4_2.csv")

# Define the file path for QAQC log file - NO Need to change just check year
file_path <- "Raw Data/SEAL_COMPASS_Synoptic_QAQC_Log_2024.csv"
final_path <- "Processed Data/COMPASS_Synoptic_Nutrients_202405.csv"

#record any notes about the run or anything other info here:
run_notes <- "Samples were put in the wrong vials, so
MSM_202405_UP_LysA_ was switched to MSM_202405_UP_LysB_,
MSM_202405_UP_LysB_ was switched to MSM_202405_UP_LysC_,
MSM_202405_UP_LysC_ was switched to MSM_202405_UP_LysA_.
It looks like NH3 was only ran for GCW, MSM, and SWH.
It looks like PO4 was only ran for GCW and MSM.
Mean NOx Reduction Efficiency is less than 95% (94.18327%).
Most of the NOx blanks are above the lower 25% quartile of samples.
3 NOx dups and one NH3 dup have high CVs, but more than 60% have low CVs.
2 NOx spikes have high CVs, but more than 60% have low CVs."

#Set up file path for metadata
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2024.csv"
```

##Setup

##Read in metadata and create similar sample IDs for matching to samples

0.1 Import Data & Clean

```
##Fix data entered incorrectly
```

```
####Fix Sample IDs####
```

```
df_all$Sample_Name[df_all$Sample_Name == Old_ID_1] <- New_ID_1
```

#Note from Zoe and Steph: It looks like we did not get any samples for MSM_202405_UP_LysA_10cm. According to the corrections on the scanned datasheets, it looks like for Upland, B samples were accidentally put in the A vials, C samples were put in B vials, and A samples were put in C vials. So, I'm thinking I can check if any analytes have MSM_202405_UP_LysA_10cm, then all the samples need to be corrected A → B, B → C and C → A.

```
## [1] "The string 'MSM_UP_202405_LysA_10cm' was found in the dataframe."
```

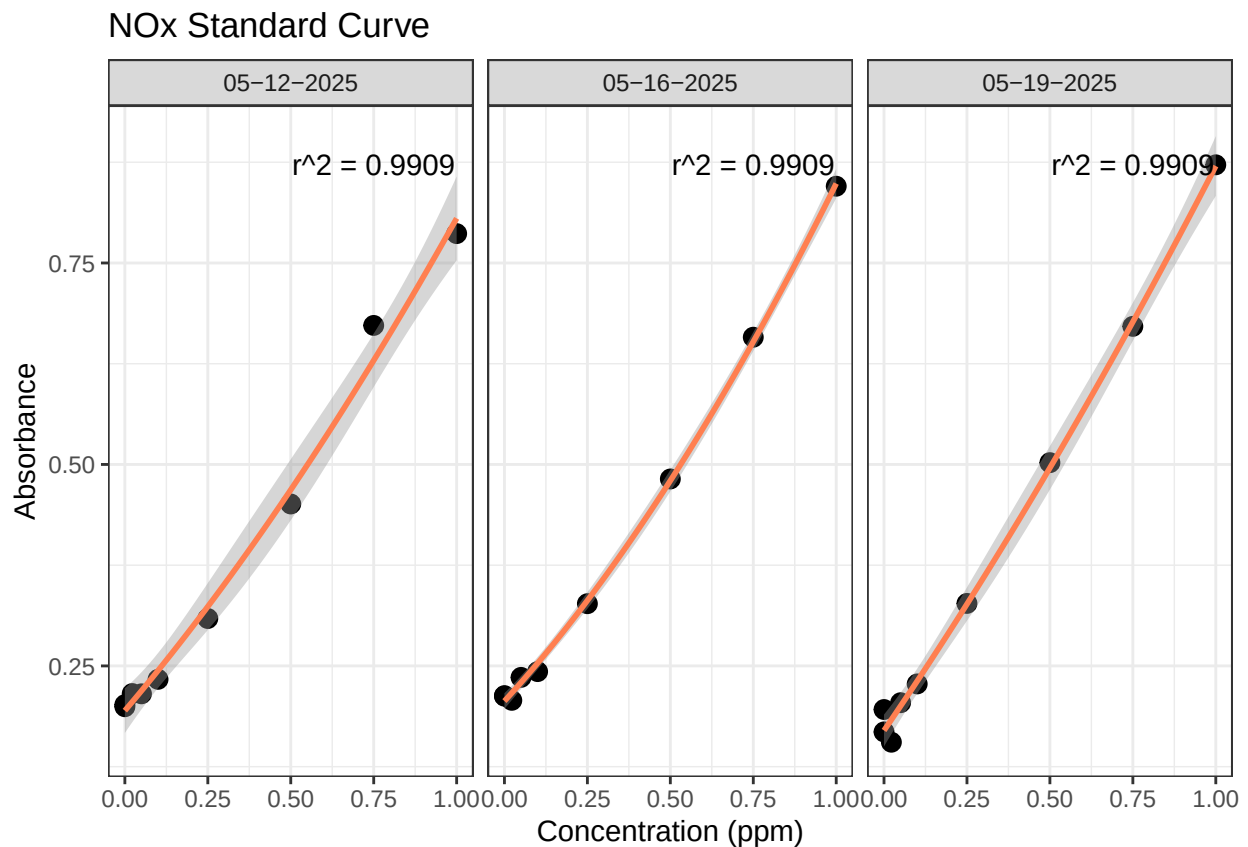
0.2 Assessing standard Curves

```
#Pull out standards data
```

```
## Assess Standard Curves
```

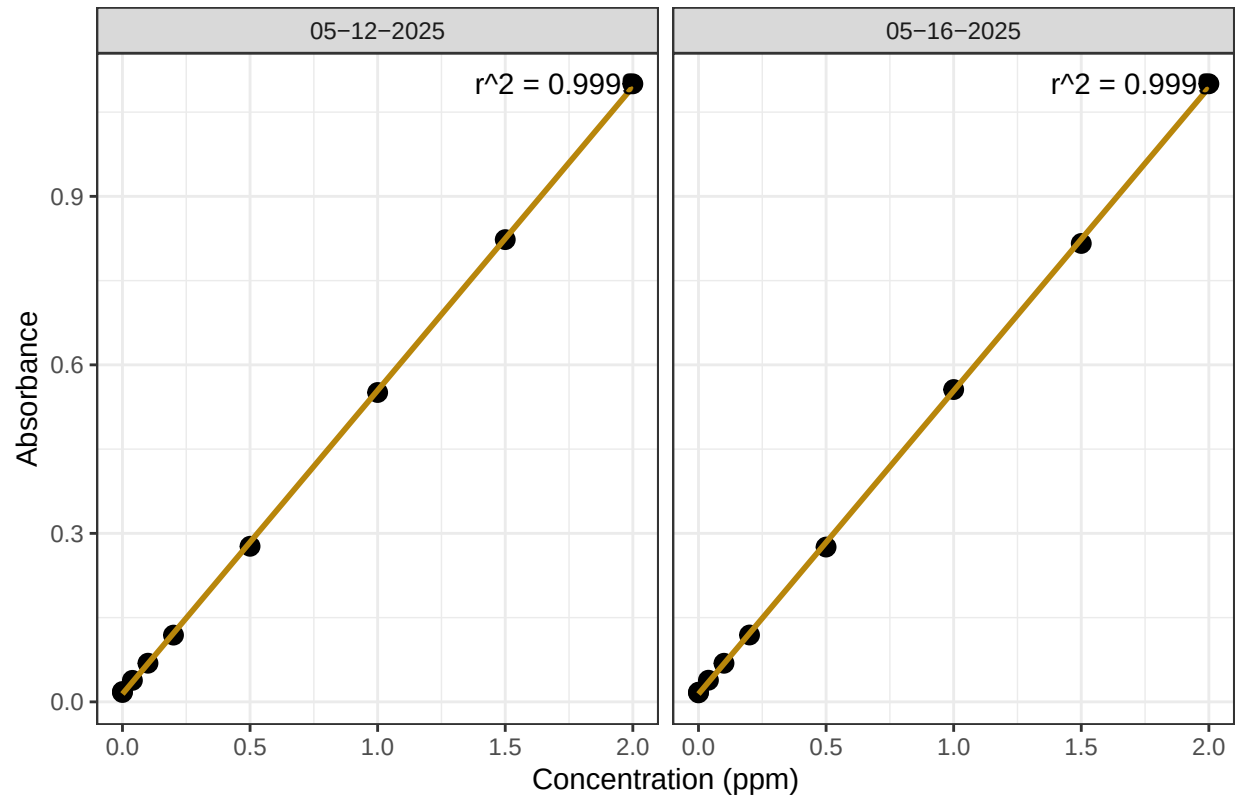
0.3 Plot standards data

```
## Assess Standard Curves
```



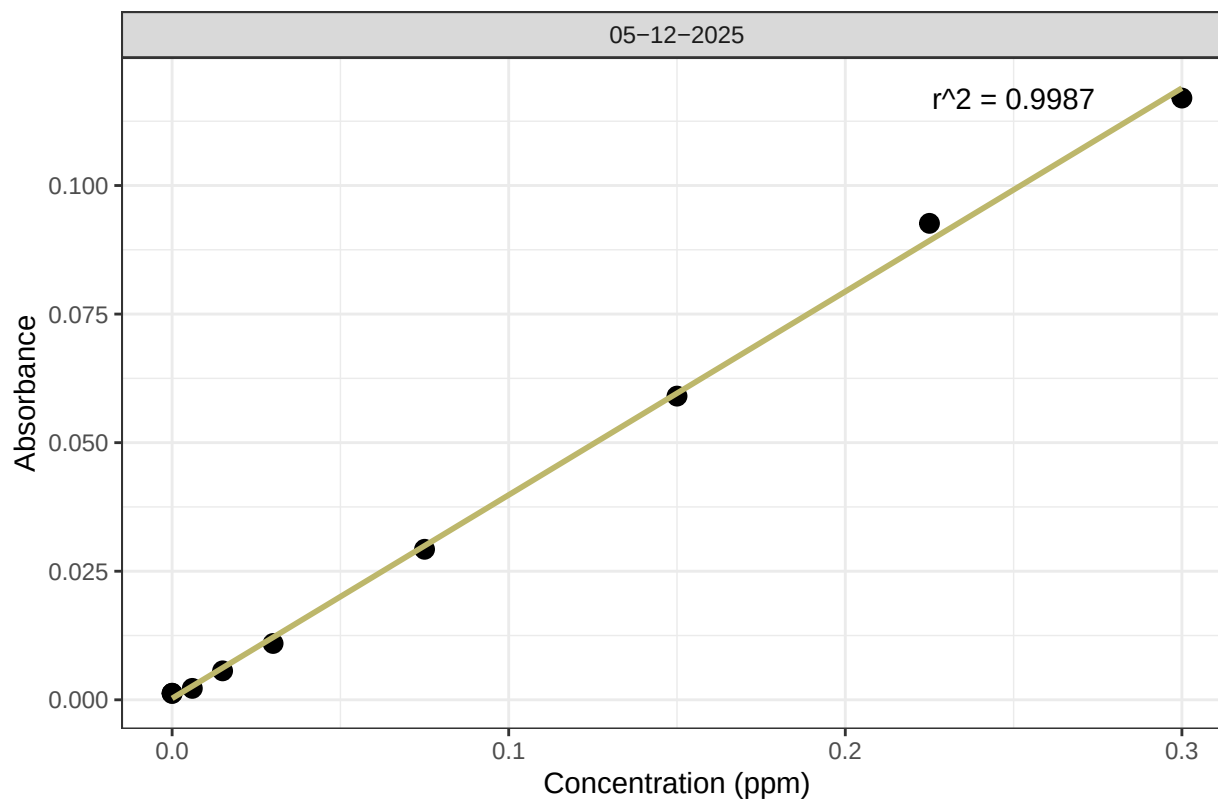
```
## 'geom_smooth()' using formula = 'y ~ x'
```

NH3 Standard Curve



```
## 'geom_smooth()' using formula = 'y ~ x'
```

PO4 Standard Curve



```
## [1] "NOx Curve r2 GOOD - PROCEED"
```

```
## [1] "NH3 Curve r2 GOOD - PROCEED"
```

```
## [1] "PO4 Curve r2 GOOD - PROCEED"
```

```
## [1] "QAQC log file exists and has been read into the code."
```

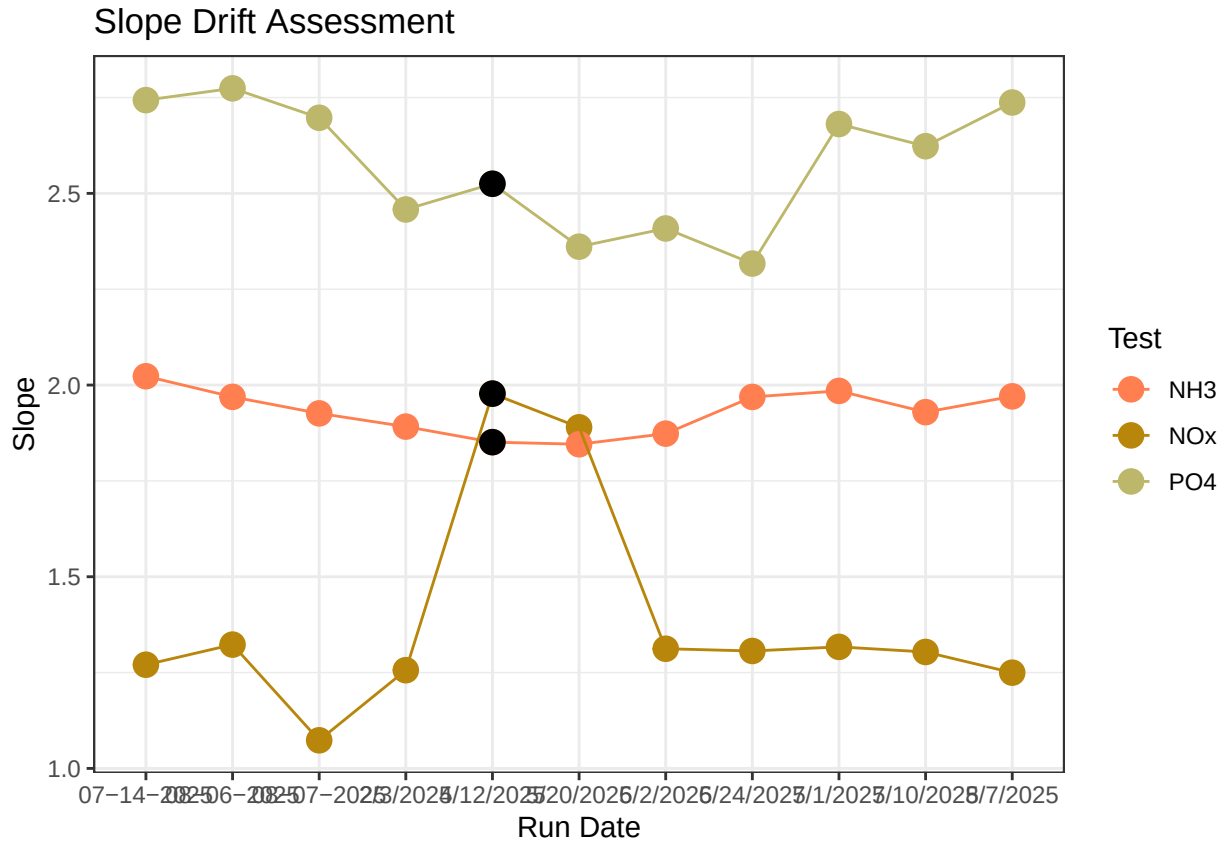


Table 1: Average Slope by Analyte

Test	avg_slope
NH3	1.930
NOx	1.389
PO4	2.575

0.4 Dilution Corrections - ensure the latest dilution is kept

```
## Dilution Corrections
```

```
## Duplicated samples: MSM_TR_202405_LysA_20cm, MSM_TR_202405_LysB_45cm, MSM_TR_202405_LysC_45cm, GCW_W
```

```
##
```

```
## All duplicated samples have valid dilutions. No naming issues detected.
```

0.5 Performance Check

```
## [1] "NOx pe Check has a % Difference <25% - PROCEED"
```

```
## Run mean = 1.575157
```

```
## Expected = 1.51
```

```
## [1] "NH3 pe Check has a % Difference <25% - PROCEED"
```

```
## Run mean = 1.037336
```

```
## Expected = 1.034
```

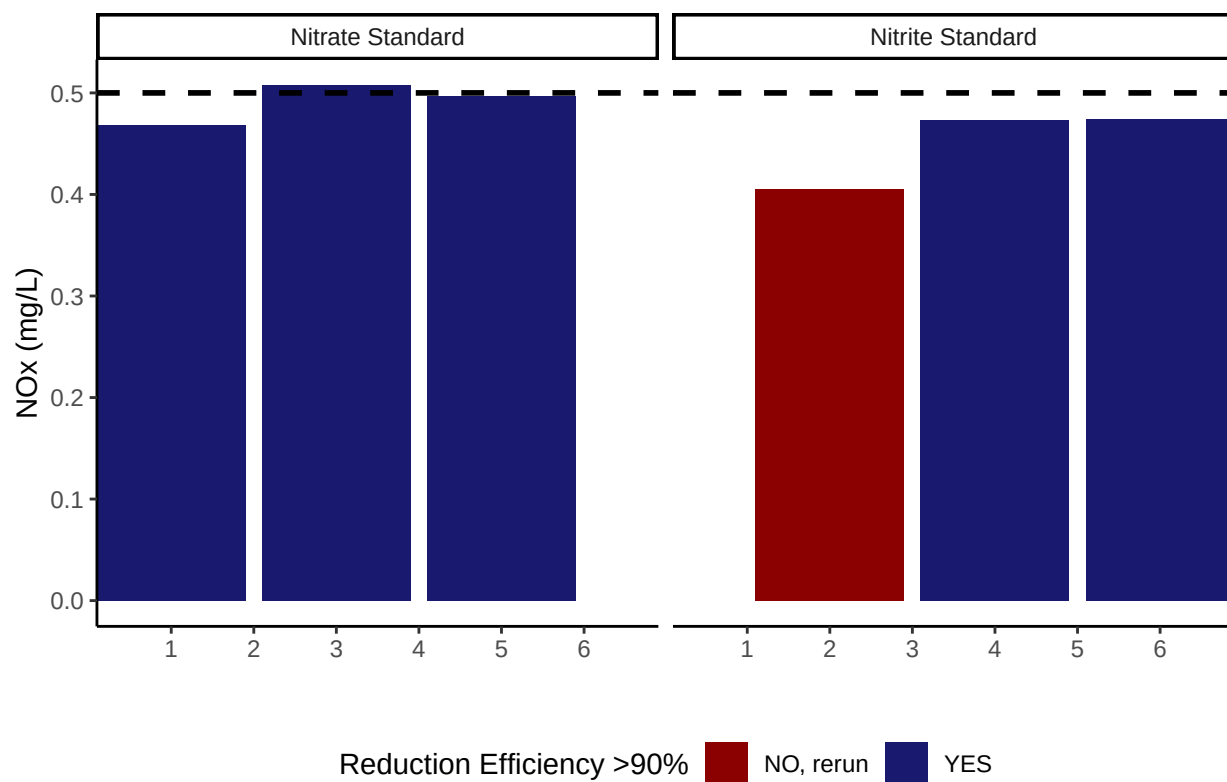
```
## [1] "P04 pe Check has a % Difference <25% - PROCEED"
```

```
## Run mean = 0.94957
```

```
## Expected = 0.824
```

```
##Check NOx Reduction Efficiency
```

```
## Assess Reduction Efficiency
```



```
## [1] "Mean NOx Reduction Efficiency <95% - REASSESS"
```

```
## [1] 94.18327
```

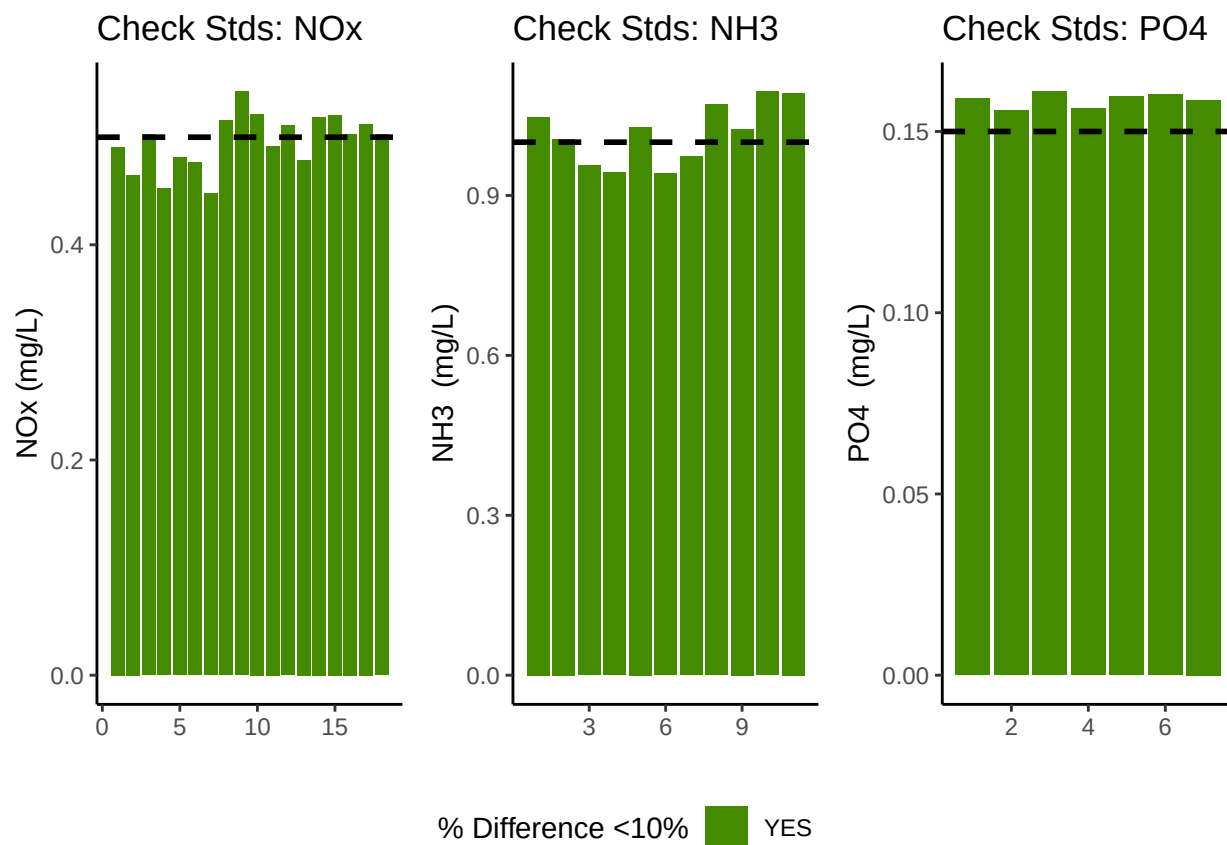

0.6 Analyze the Check Standards

```
## Analyze Check Standards
```

```
## [1] "NOx Check Standard RSD within Range - PROCEED"
```

```
## [1] "NH3 Check Standard RSD within Range - PROCEED"
```

```
## [1] "PO4 Check Standard RSD within Range - PROCEED"
```



```
## [1] ">60% of NOx Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">60% of NH3 Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">60% of PO4 Check Standards are within range of expected concentration - PROCEED"
```

0.7 Analyze Blanks

```
## Assess Blanks
```

```
## [1] "<60% of NOx Blanks are below the lower 25% quartile of samples or 1/2 the detection limit - REAS
```

```
## [1] ">60% of NH3 Blanks are below the lower 25% quartile of samples - PROCEED"
```

```
## [1] ">60% of PO4 Blanks are below the lower 25% quartile of samples- PROCEED"
```

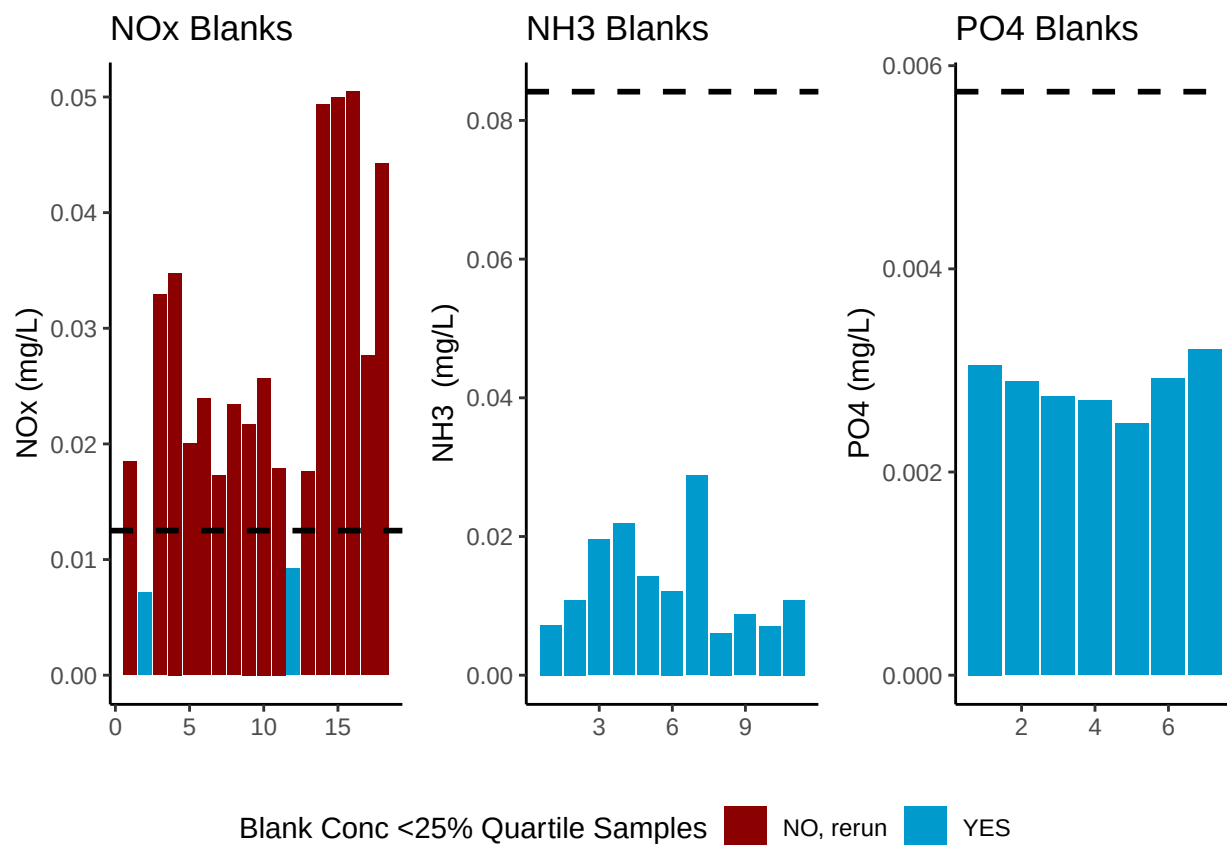


Table 2: Mean Concentration of Blanks

Test	Blank_Mean_Conc
NOx	0.0273
NH3	0.0134
PO4	0.0029

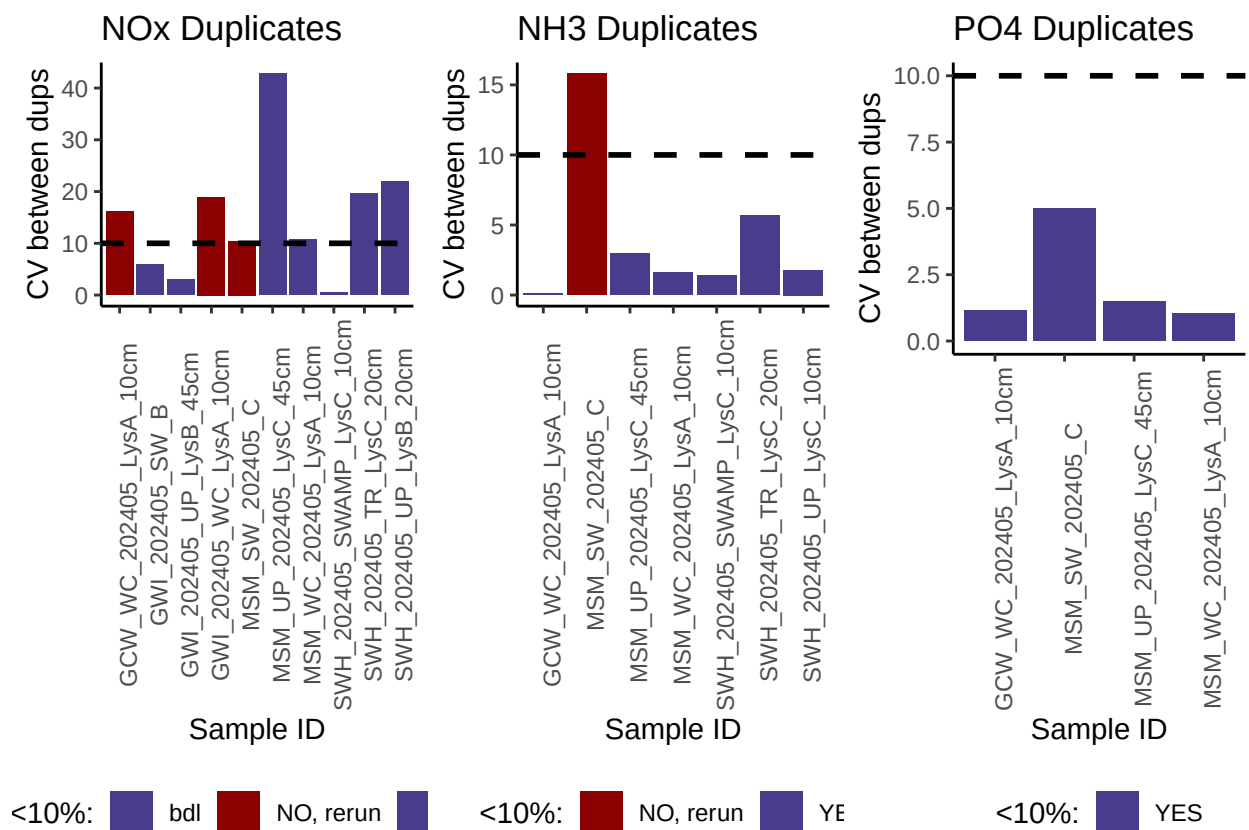
0.8 Analyze Duplicates

```
## Analyze Duplicates
```

```
## [1] ">60% of NOx Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">60% of NH3 Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">60% of PO4 Duplicates have a CV <10% - PROCEED"
```



0.9 Spikes

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

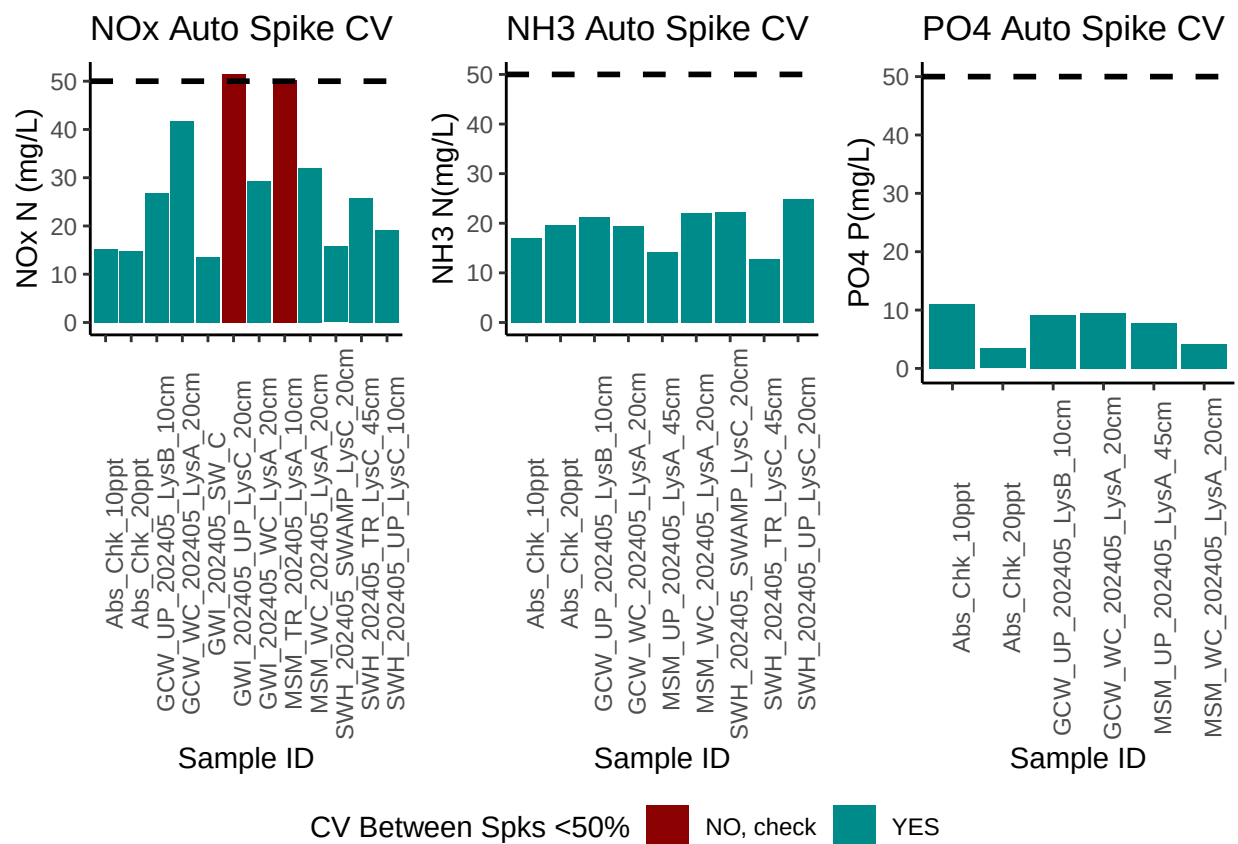
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
```

```
## i Please use 'linewidth' instead.
```

```
## This warning is displayed once every 8 hours.
```

```
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
```

```
## generated.
```



0.10 Matrix Effects

```
## [1] "NO NOx Matrix Effect, PROCEED"
```

```
## [1] "NO NH3 Matrix Effect, PROCEED"
```

```
## [1] "NO PO4 Matrix Effect, PROCEED"
```

0.11 Unit Converted Data Column Added (mg/L to uM)

0.12 Sample Flagging - Within range of standard curve

```
## Sample Flagging
```

0.13 Pull out sample id information

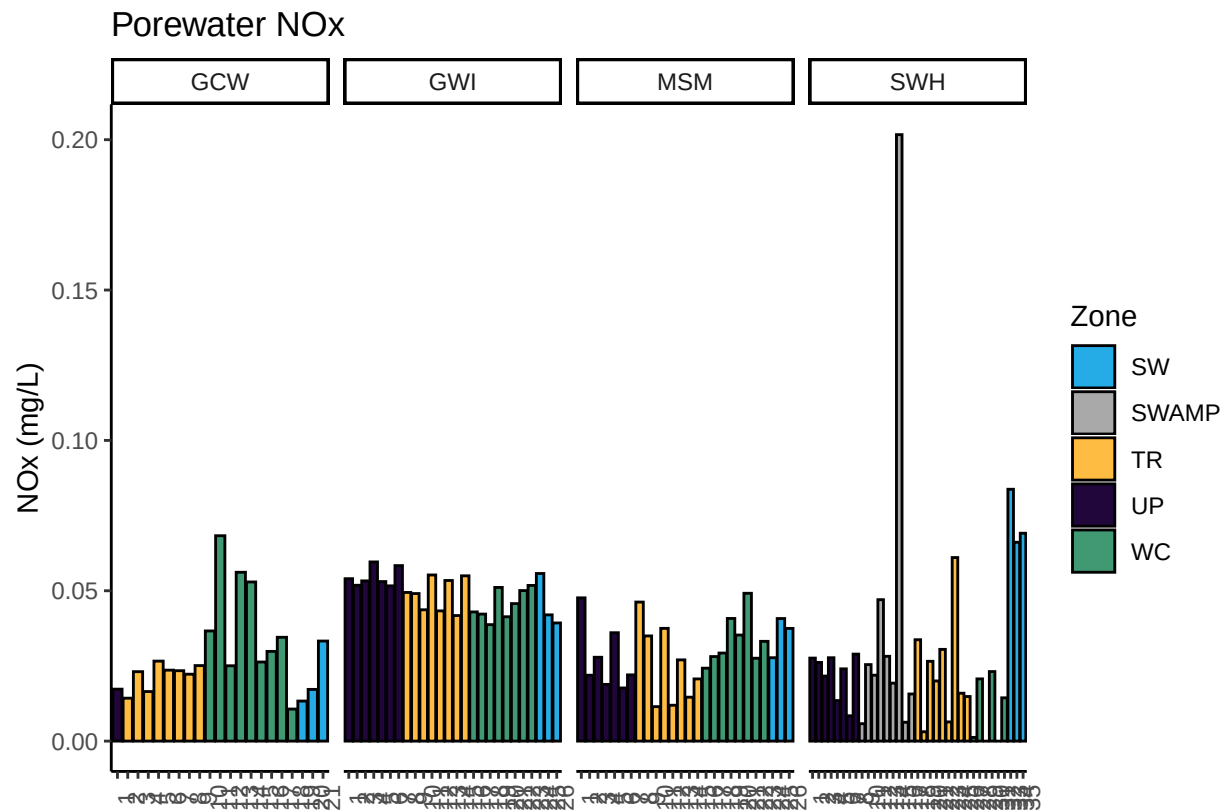
```
## Sample Processing
```

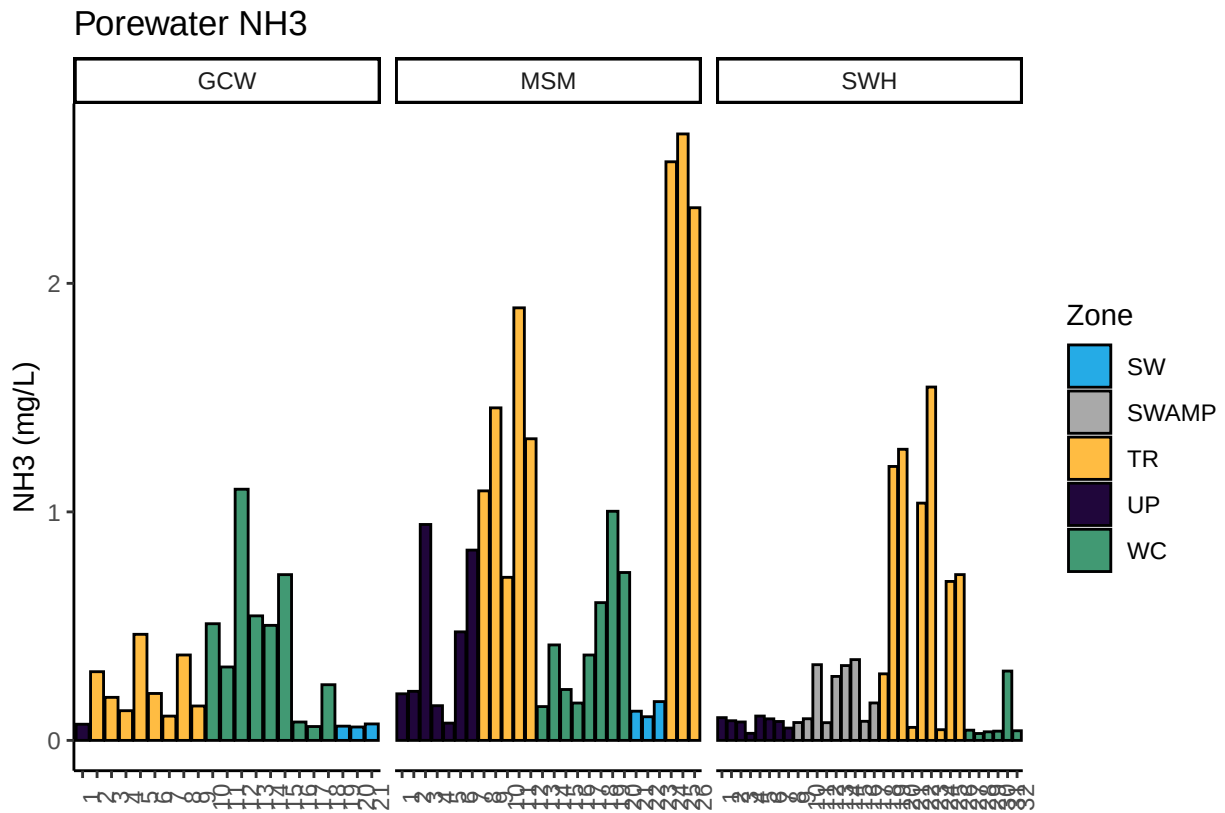
0.14 Check to see if samples run match metadata & merge info

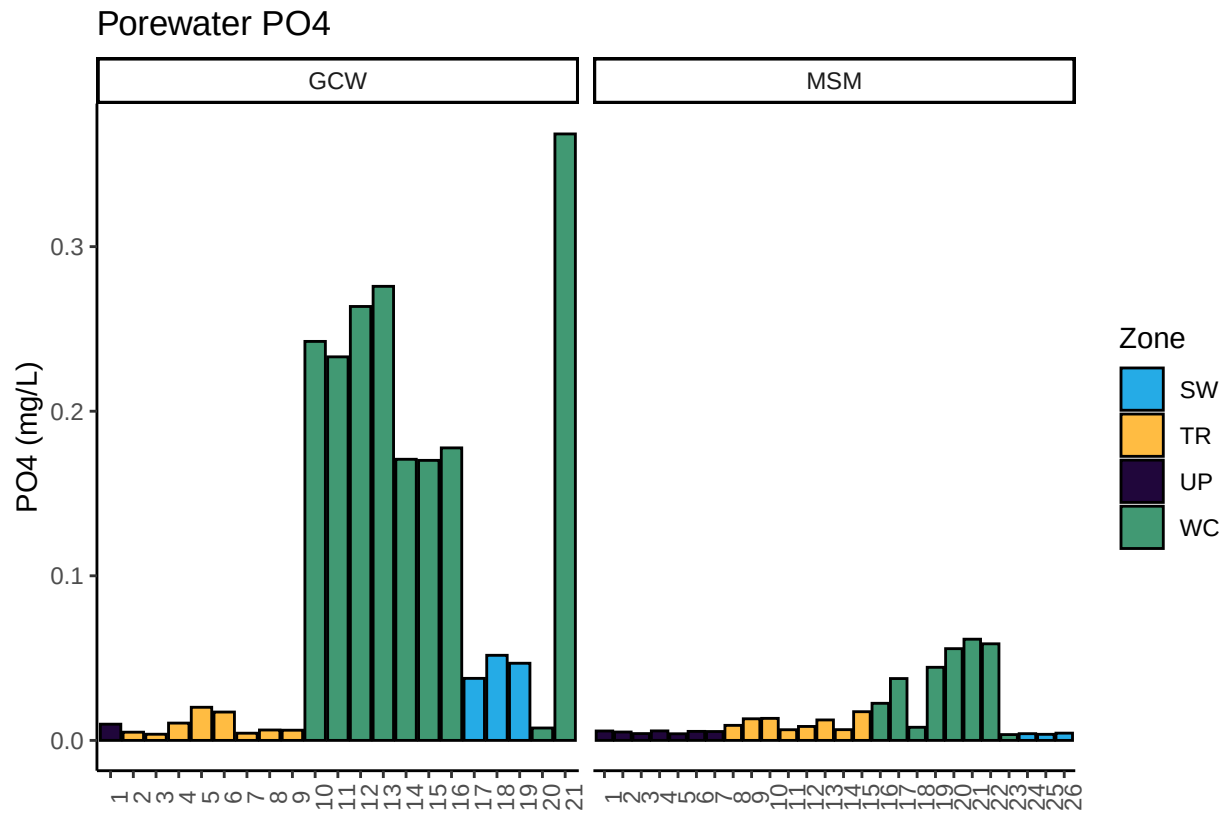
```
## Check Sample IDs with Metadata
```

```
## All sample IDs are present in metadata.
```

Visualize Data







0.16 Export Processed Data

#end